cribo

V, 3 elementary vec; all =0, for solving → find

Section 6.4 Finite Dimensional Spaces

set of vec that rep entire space s/t

Exercise 6.4.1

In each case, find a basis for V that includes the vector v.

c. $V = M_{22}$, $v = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow$ Basis vectors can be a combo of all four elementary vectors

$a\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + b\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Then to see if valid; check for linear independence s/t $a=b=c=d=0$

$TL \Rightarrow a=0$
$TR\ a+b=0$
$bL \Rightarrow a+c=0$
$bR \Rightarrow a+d=0$

proving our set/Basis is valid!

Hence Basis 'B' $= \{v, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$; We could also have used $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ for one of the open vectors.

Exercise 6.4.2

In each case, find a basis for V among the given vectors.

b. $V = P_2$, $\{x^2 + 3, x + 2, x^2 - 2x - 1, x^2 + x\}$

$\cancel{\boxed{\begin{matrix} x^2+3 \\ x+2 \\ x^2-2x-1 \\ x^2+x \end{matrix}}}$ dim = 4

$|\text{Basis}| = \dim(A)$

if more vec than dim; set dependent
else fewer; can't span

polynom. deg $2 \leq 0$

Then it rep $\{1, x, x^2\}$; dim $= 3 \neq 4$?

① - Try first 3 vectors for basis s/t

$a(x^2+3) + b(x+2) + c(x^2-2x-1)$

$= x^2(a+c) + x(b\cancel{x}-2c) + (3a+2b-c)$

$= x^2(a+c) + x(b-2c) + 1(3a+2b-c)$

⑤ Treat each as ind = 0 s/t

$a+c=0 \Rightarrow a=-c$
$b-2c=0$ $\cancel{c=b/2}$ $b=2c$
$3a+2b-c=0 \Rightarrow \cancel{\frac{3b}{2}+2b-\frac{b}{2}} = 0$

$\Rightarrow 3c+4c$ ...
$-0=0$ ...

Try with $v_3 = v_4$ instead (swap)

$\Rightarrow a(x^2+3) + b(x+2) + c(x^2+x)$

$x^2(a+c) + x(b+c) + 1(3a+2b)$

$a=c$, $c=b$ $\Rightarrow 3b+2b = 5b \neq 0$

$\Rightarrow b=0;\ a=b=c=0$

Meaning the set we found is linearly ind! which make basis;

$B = \{x^2+3, x+2, x^2+x\}$

Exercise 6.4.5

In each case use Theorem 6.4.4 to decide if S is a basis of V.

b. $V = P_3$; $S = \{2x^2, 1 + x, 3, 1 + x + x^2 + x^3\}$

dim must be 4 $\{1, x, x^2, x^3\}$

dim is $|S|$ so can proceed for test;

$a(2x^2)+b(1+x)+c(3)+d(x^3+x^2+x+1)$

$x^2(2a+d)$, $x(b+d)$, $1(b+d)$, $x^3(d)$

$2a=0$, $c=0$, $b=0$, $d=0$; $a=b=c=d=0$ $\Rightarrow$ is a basis and ~~lin. ind~~ linearly ind. $\therefore B = S$

Test Test

Section 7.1 Examples and Elementary Properties

Exercise 7.1.1

Show that each of the following functions is a $\underbrace{\text{linear transformation}}_{\text{See think of}}$. 1) 0-vec 2) + 3) r

g. $T : P_n \to \mathbb{R}$; $T(r_0 + r_1x + \cdots + r_nx^n) = r_n$

To satisfy 0-vector; note that b/c $P_n \to \mathbb{R}$, ~~$\mathbb{R} \in 0$~~ $\mathbb{R}$ contains '0'; the transformation for $x^n = 0$, $r_0 = 0 \Rightarrow 0$ showing 0-vector holds

2) for addition; just show that splitting the transformation holds true such that $r_0 + r_1x + \ldots = $ 'a' $\ldots + r_nx^n + r_{n+1}x^{n+1} = $ 'b' then $a+b$ for $T(a+b) = a+b$ as it's the same transformation.

an easy way to 'break' this is if we had $T(r_n + \ldots)^{1/2}$ would break it

3) no constants, exponents (outside) or abs values $\Rightarrow$ adding some const const 'c' $\in \mathbb{R}$ then $cT(r_0 + \ldots) = T(cr_0 + c\ldots)$ rule of scalar multiplication holds true

Test!
Test!
TestTestTestTest

already know it's a linear T; just proving it tho

literally says; n^{th} dim $\mathbb{R}$ * basis of that dim = transformation

j. $T : \mathbb{R}^n \to V$; $T(r_1, \cdots, r_n) = r_1e_1 + \cdots + r_ne_n$ where $\{e_1, \ldots, e_n\}$ is a fixed basis of V

1) 0-vector; $0 \times c = 0 \Rightarrow$ must hold true even when multiplied by basis vector $\Rightarrow$ 0-vector holding true

2) addition is the same; b/c there's no constants or exponents such as r_n^t; we can split our transformation in half s/t a is first half and b is second half, then $a+b = T(a+b) = T(r_1 + \cdots + r_n)$

3) Since our Transformation is pretty much the same $c r_n$, $c \in \mathbb{R}$; it's just a matter of formally stating what's happening s/t $cT(r_1 + r_n) = T(cr_1 + cr_n)$ we see it still holds as its just adding another constant to the basis.

'x'

Exercise 7.1.4

power!

In each case, find a linear transformation with the given properties and compute $T(v)$.

a. $T : \mathbb{R}^2 \to \mathbb{R}^3$; $T(1, 2) = (1, 0, 1)$, $T(-1, 0) = (0, 1, 1)$; $v = (2, 1)$

2 transformations:

$T(1,2) \Rightarrow (1,0,1)$, Then $\vec{v}(2,1) = s(1,2) + t(-1,0)$

$T(-1,0) = (0,1,1)$

such that we form for solving a SOLE, then solving it we get

$1s - 1t = 2 \Rightarrow 2s = 4/2$

$2s = 0$

$t = -\frac{3}{2}$

$\Rightarrow \frac{1}{2}(1,0,1) + (-\frac{3}{2})[0,1,1]$

$= \boxed{\left(\frac{1}{2}, -\frac{3}{2}, -1\right)}$

as final answer!

decompose transformations; via properties of linear transformations

Exercise 7.1.5

If $T : V \to V$ is a linear transformation, find $T(v)$ and $T(w)$ if:

a. $T(v+w) = v - 2w$ and $T(2v - w) = 2v$

Have 2 eq;

$T(v)+T(w) = v-2w$ ①

$T(2v)-T(w) = 2v$ ②

$\Rightarrow$ ∴ ①+② $\Rightarrow (T(v)+T(w)) + (T(2v)-T(w)) = (v-2w)+2v$

$\Rightarrow 3T(v) = 3v-2w$; sub into ①

$T(v) = v - \frac{2}{3}w$

$\Rightarrow v - \frac{2}{3}w + T(w) = v-2w$

$T(w) = -2w + \frac{2}{3}w$

$= -\frac{4}{3}w$

$T(v) = v - \frac{2}{3}w$

$T(w) = -\frac{4}{3}w$

//

Section 7.2 Kernel and Image of a Linear Transformation

Exercise 7.2.2

In each case, (i) find a basis of $\ker T$, and (ii) find a basis of $\operatorname{im} T$. You may assume that T is linear.

assuming means all polynomial deg $2 \leq 0$

b. $T : P_2 \to \mathbb{R}^2$; $T(p(x)) = (p(0), p(1))$

Basis(ker).

• $p(0) \Rightarrow a = 0$

• $p(1) \Rightarrow a+b+c = 0$

$b+c=0$

$b=-c$

$\Rightarrow 0=0$ which means the entire space is valid $\Rightarrow b, c \in \mathbb{R}$

Then must take form for $a + bx^1 + cx^2$; x^0 can be dropped implied

$\Rightarrow bx^1 + cx^2$

~~Basis~~ $\ker(T) = \{bx + cx^2\}$

ker T

Basis $\ker T = \{bx + cx^2;\ b, c \in \mathbb{R}\}$ //

Image; Know $\dim(P_2) = 3$

$\dim \ker = 1$

Then $3 = 1 + a$

$a = 2$;

any standard basis vec in $\mathbb{R}^2$ will work s/t

$\operatorname{Im} T = B = [\{1,0\},\{0,1\}]$ //

4

e. $T : M_{22} \to M_{22};\ T\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+b & b+c \\ c+d & d+a \end{bmatrix}$ = 'P'

Since given actual vector this time; 1) check all = 0 consistent,

$$\left.\begin{aligned} a+b&=0 \\ b+c&=0 \\ c+d&=0 \\ d+a&=0 \end{aligned}\right\} \quad \begin{aligned} a&=-b \\ b&=-c \\ c&=d \\ b+-b&=0 \Rightarrow 0=0! \end{aligned}$$

Then expressing all in terms of ~~a scalar 't'~~ b

$\Rightarrow b\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$ which gives us our working vector $= \ker(T)$ //

To find Basis, we simply let P for each var = 1, rest = 0 s/t a=0, b=1, rest=0 ... each=1 giving us 4 possible vectors

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$; Since dim = 3, we can chose any of these 3 for example $B = \left\{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right\}$ //

Section 7.3 Isomorphisms and Composition

Exercise 7.3.1

Verify that each of the following is an isomorphism (Theorem 7.3.3 is useful).

a. $T : \mathbb{R}^3 \to \mathbb{R}^3;\ T(x, y, z) = (x + y, y + z, z + x)$

First need to find $\ker(T) \Rightarrow$

$$\left.\begin{aligned} x+y&=0 \\ y+z&=0 \\ x+z&=0 \end{aligned}\right\} \left.\begin{aligned} x&=-y \\ z&=-y \end{aligned}\right\} \Rightarrow -2y=0,\ y=0=x=z \Big\} \therefore \ker T = \{(0,0,0)\} //$$

dim of T = 3 from $\mathbb{R}^3$

$\therefore$ Transformation is 1-1 s/t dim(V) = dim(W) = 3

$\therefore$ it is isomorphism since one-to-one and onto $\Rightarrow$ producing the isomorphic set.

h. $T : M_{mn} \to M_{nm}$; $T(A) = A^T$

for $T(A)=0$; $A^T=0$ implying $A=0$ $\ker T = 0$ fulfills $1 \to 1$ relationship

for onto $\Rightarrow A = B^T$ s/t $T(A) = (B^T)^T = B$ ← By rules for set operations ~~transformations~~

Both are same ~~size of~~ ~~M~~ dimension for mn, nm

$\Rightarrow$ Thus linear $1 \to 1$ onto $\Rightarrow$ isomorphism

Exercise 7.3.4

Remember it's 'T' then S!!!

In each case, compute the action of ST and TS, and show that $ST \neq TS$.

d. $S : M_{22} \to M_{22}$ with $S \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$; $T : M_{22} \to M_{22}$ with $T \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & a \\ d & b \end{bmatrix}$

for 'ST' we apply T first $\Rightarrow \begin{bmatrix} c & a \\ d & b \end{bmatrix} S \Rightarrow \begin{bmatrix} c & 0 \\ 0 & b \end{bmatrix}$ as our 'ST'

Then for 'TS' we get

$\Rightarrow \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} T \Rightarrow \begin{bmatrix} 0 & a \\ d & 0 \end{bmatrix} \neq \begin{bmatrix} c & 0 \\ 0 & b \end{bmatrix}$ thus showing they aren't equal.

Exercise 7.3.6

Determine whether each of the following transformations T has an inverse and, if so, determine the action of T^{-1}.

a. $T : \mathbb{R}^3 \to \mathbb{R}^3$; $T(x, y, z) = (x + y, y + z, z + x)$

In order to have an inverse, require transformation to be 1→1

$\Rightarrow$ To check already know $T(0,0,0) \Rightarrow 0 \therefore T^{-1}(0,0,0) \Rightarrow 0 = T(0) \therefore$ has A inverse

To determine action;

$$\left.\begin{array}{l} x+y=a \\ y+z=b \\ z+x=c \end{array}\right\} \Rightarrow \begin{array}{l} x-z=a-b \\ a-b+c \\ x=\frac{1}{2}(a-b+c) \\ y=a-\frac{1}{2}(a-b+c)=\frac{1}{2}(a+b+c) \\ z=\frac{1}{2}(a+b+c) \end{array}$$

$\Rightarrow$ put into $T^{-1}(x,y,z)$

$\Rightarrow \underbrace{T^{-1}(\cdot) = \frac{1}{2}(a-b+c, a+b-c, -a+b+c)}_{\text{as our answer}}$

where each show $a,b,c \Rightarrow x\,y\,z$ in inverse plane